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Let's denote E_k to be the event that after k -th draw the numbers form an interval, $\forall k=1, 2, \dots, n$. [Let's denote interval $[i, i+1, \dots, i+k]$ by $[i, i+k]$]

$P(E_1) = 1$, trivially (Since the first draw always results in an interval)

$$P(E_2 | E_1) = \left(\frac{1}{n-1}\right) \cdot \left(\frac{2}{n}\right) + \left(\frac{2}{n-1}\right) \cdot \left(\frac{n-2}{n}\right) = \frac{2+2n-4}{n(n-1)} = \frac{2(n-1)}{n(n-1)} = \frac{2}{n}$$

Exactly one choice for second draw, namely 2 or $n-1$ respectively

First draw is 1 or n

Two choices for the second draw, namely $i-1$ and $i+1$ if first draw is i where $i \neq 1$ and $i \neq n$

At k th draw one can choose only $i-1$ or $i+k-1$ if $(k-1)$ th draw forms the interval $[i, i+k-2]$, $k \geq 3$

$$P(E_3 | E_2 E_1) = \left(\frac{1}{n-2}\right) \cdot \left(\frac{2}{n-1}\right) + \left(\frac{2}{n-2}\right) \cdot \left(\frac{n-3}{n-1}\right) = \frac{2(n-2)}{(n-1)(n-2)}$$

Exactly one choice for third draw

First two draws form interval $[1, 2]$ or $[n-1, n]$

Exactly 2 choices for the third draw, namely, $i-1$ and $i+2$ if the first two draws form interval $[i, i+1]$

First two draws form an interval which is different from $[1, 2]$ and $[n-1, n]$ there are $n-1$ such intervals

In general, $P(E_k | E_{k-1} E_{k-2} \dots E_1)$

$$= \left(\frac{1}{n-k+1}\right) \cdot \left(\frac{2}{n-k+2}\right) + \left(\frac{2}{n-k+1}\right) \cdot \left(\frac{n-k}{n-k+2}\right)$$

$$= \frac{2(n-k+1)}{(n-k+1)(n-k+2)} = \frac{2}{n-k+2}, \text{ for } 2 \leq k \leq n$$

$$\therefore P(E_n) = \prod_{k=2}^n P(E_k | E_{k-1} E_{k-2} \dots E_1) \cdot P(E_1) = \prod_{k=2}^n \left(\frac{2}{n-k+2}\right) \cdot (1)$$

$$= \frac{2}{n} \cdot \frac{2}{n-1} \cdot \frac{2}{n-2} \dots \frac{2}{2} = \frac{2^{n-1}}{n!} \quad (\text{Ans})$$